Multimodal Transformers for Land-Cover Distribution Prediction

Ilayda Ergür
Middle East Technical University
Ankara, Türkiye
ilayda@ergur.org

***Abstract*—Transformer-based models have enabled effective multimodal learning by combining visual and textual information. This study investigates the research question: *Does integrating textual descriptions improve the prediction of land-cover distributions derived from segmentation masks in remote sensing images?* The problem is defined as a multi-output regression problem. It proposes a multimodal transformer model that consists of a Vision Transformer and DistilBERT. Early findings indicate that the multimodal model works much better than an image-only baseline, which suggests that textual information is a complementary source of semantic information.**

***Index Terms*—multimodal learning, transformers, remote sensing**

I. Introduction

The architectures based on transformers have enhanced visual and multimodal learning significantly. The land-cover analysis in remote sensing is generally carried out only on the basis of the image data and the textual descriptions are frequently neglected.

It is examined in this study whether textual information will enhance prediction of land-cover distributions based on segmentation masks. The dataset will include images, masks, and captions, with percentages of classes as ground truth. The formulation of the task is a multi-output regression problem.

II. Literature Survey

Transformer-based models, including Vision Transformers (ViT), have been found to be high performers when it comes to modeling global image dependencies. CLIP is used in multimodal learning where the visual and textual representations are correlated with contrastive learning, which makes it possible to understand each other cross-modally. Cross modality attention extensions like CLIPSeg and LAVT add textual guidance to visual tasks. GroupViT, likewise, demonstrates that textual supervision has the ability to affect visual grouping and representation learning.

The recent development of multimodal foundation models further emphasizes the need of unified vision-language systems. Nevertheless, the majority of the available literature is dedicated to segmentation, retrieval or grounding tasks. The importance of textual information to solve structured prediction problems, including predicting land-cover distributions given segmentation masks, is somewhat understudied. This is the gap that drives the approach proposed.

III. Proposed Method

The suggested system relies on a multimodal transformer-based system. The visual features of images are obtained with the help of a Vision Transformer, and the textual descriptions are encoded into semantic embeddings with the help of a DistilBERT model. The combination of these representations is done by feature concatenation to create a combined multimodal representation.

The regression head is used to predict seven values of land-cover composition. A model that is based on image-only transformers is utilized as a baseline. The multimodal model builds up on this foundation by adding textual information. The assessment is conducted on the basis of Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

IV. Proof of Concept

Preliminary experiments were done in order to determine feasibility. A stratified 70/15/15 split was made according to the predominant class.

Two models were used: a pure image Vision Transformer and a multimodal model that uses the combination of Vision Transformer and DistilBERT.The results are shown in Table I.

TABLE I
PRELIMINARY RESULTS

Model	V-MAE	V-RMSE	T-MAE	T-RMSE
ViT	4.34	8.78	3.32	8.65
ViT + DistilBERT	**2.61**	**4.93**	**2.64**	**4.83**

The multimodal model significantly improves performance, reducing MAE by approximately 35% and RMSE by over 40%. The consistency between validation and test results indicates good generalization. These findings suggest that textual information provides complementary semantic cues that enhance prediction accuracy.

V. Conclusion

This paper indicates that incorporation of textual information enhances prediction of land-cover distributions in remote sensing images. The suggested multimodal transformer-based system is more effective than the image-only reference, which confirms the efficiency of visual and textual modalities integration. Future directions will involve the investigation of attention-based fusion approaches and in-depth ablation experiments to increasingly understand the role of modality.

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